ORAL · 3RD PLACE · AKBC 2026 · LM-KBC SHARED TASK

EMNLP2026 BUDAPEST

# The Instruction Tax on Factual Recall

## Base Models Can Beat Their Instruction-Tuned Counterparts

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## 1 AKBC Shared Task

### Closed-book knowledge base construction

**Predict all valid objects for a subject–relation pair.**

- **Closed-book:** no retrieval, no external data, no fine-tuning; under 32B parameters.
- **Six relations:** 477 / 475 / 475 train / val / test.
- **Nearly 20% unanswerable:** abstention is scored.

### Instruction tax

> Instruction tuning decreases factual recall. The cause is the **weights**, not the prompt.

hasArea probe, gemma-3-27b:
base **0.880**, tuned **0.590**.

## 2 Model Selection

### Completion prompt

Base model. No chat template, no instruction.

> The total area of Lake Victoria is 68800 square kilometres.
> The total area of Bodensee is 536 square kilometres.
> **The total area of Vygozero is** ▮

One exemplar per quintile of hasArea values. Set relations: about one third of exemplars are “none”.

ONE FORMAT PER RELATION

| |
|---|
| The seating capacity of {s} is {v}. |
| Countries sharing a land border with {s}: {v} |
| {s} stock exchange listings: {v} |
| Recipients of {s}: {v} |

### Knowledge probe

| FAMILY | BASE, COMPL. | IT, CHAT | IT, COMPL. | TAX Δ | CEILING |
|---|---|---|---|---|---|
| gemma-3-12b | **.720** | .370 | .320 | +.350 | .84 → .47 |
| **gemma-3-27b** | **.880** | **.590** | **.580** | **+.290** | **.95 → .69** |
| gemma-2-27b | **.720** | .640 | .640 | +.080 | .82 → .77 |
| Mistral-Nemo-2407 | **.710** | .630 | .660 | +.080 | .90 → .87 |
| Mistral-Small-3.2-24B | – | .720 | .690 | – | – |
| Magistral-Small-2506 | – | .610 | .570 | – | – |
| Qwen3-30B-A3B (MoE) | – | .430 | .430 | – | – |
| Qwen3-30B-A3B-Think. | – | .430 | .410 | – | – |

Voted accuracy, 100 hasArea validation rows. Δ = base − chat. Ceiling: correct in ≥ 1 sample.

> Same prompt, gemma-3-27b-it: **0.580**. The tuned model forgot facts.

## 3 Methodology

STEP 1

### Probe first

~100 known facts, several phrasings, 8–12 samples each.

**~1 GPU-hour per model version.**

STEP 2

### Never ask a question

Training exemplars, then an unfinished line. No chat template.

**+0.116 F1 over chat, same weights.**

STEP 3

### Keep what repeats

64–256 samples; keep answers above a fitted threshold.

**0.6218 (n = 8) → 0.6604 (n = 64).**

STEP 4

### Verify rare candidates

awardWonBy: verify low-support candidates (AUC 0.86).

**+0.027 on awardWonBy.**

STEP 5

### Dead or alive?

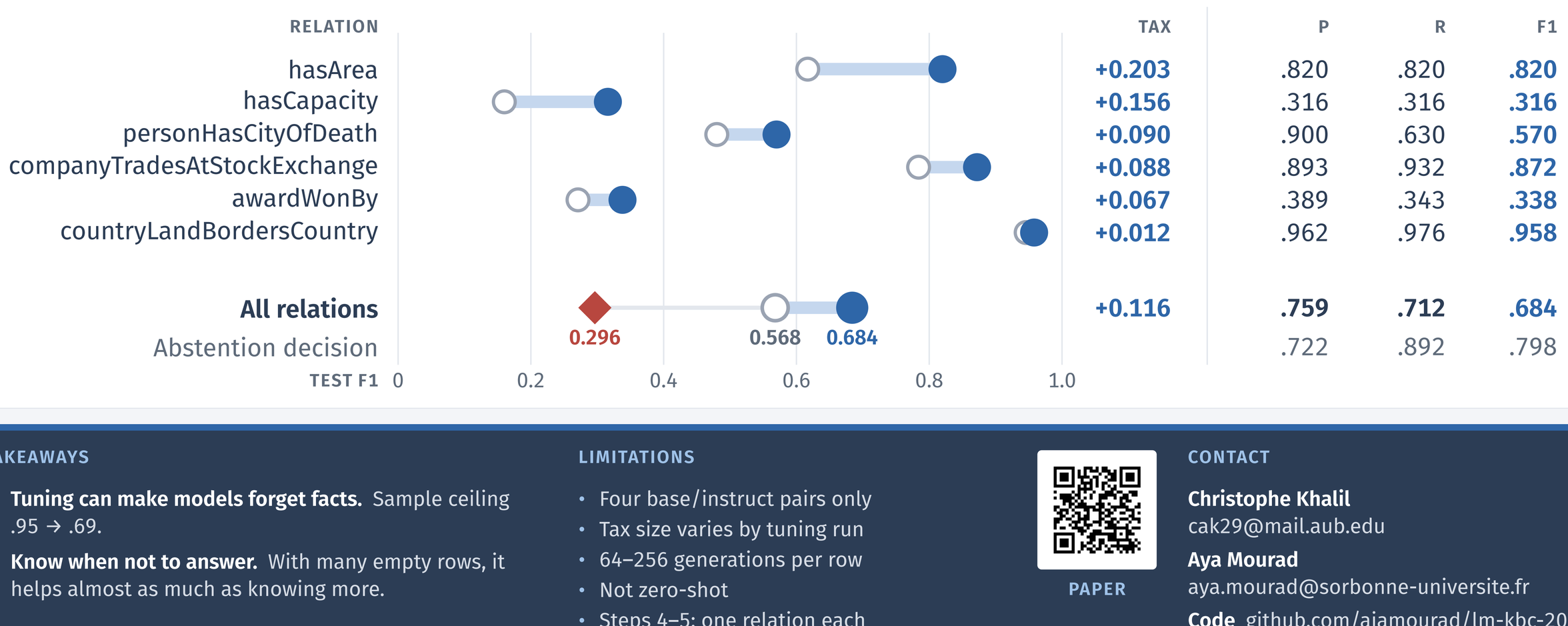
All 16 samples say alive → answer empty.

**+0.030 on death; abstention F1 0.773 → 0.798.**

## 4 Results

Official test split · gemma-3-27b-pt (27.43B) · vLLM on two H100s

### Instruction tax at full-system scale




TAKEAWAYS

1. **Tuning can make models forget facts.** Sample ceiling .95 → .69.
2. **Know when not to answer.** With many empty rows, it helps almost as much as knowing more.

LIMITATIONS

- Four base/instruct pairs only
- Tax size varies by tuning run
- 64–256 generations per row
- Not zero-shot
- Steps 4–5: one relation each

PAPER

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